
High-Resolution Mass Spectrometry Data Processing
Software for Glycan Identification
(MS-GDF Version 1.0)

User Manual

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Chapter 1 Overview of MS-GDF Software

1.1 Introduction

Glycans participate in many key biological processes, including cell adhesion, receptor activation, signal transduction, immune regulation, and disease progression. Accurate characterization of glycan structures and dynamic changes is essential for elucidating the molecular mechanisms of disease, discovering novel biomarkers, and developing targeted therapeutic strategies. However, comprehensive glycan characterization remains highly challenging because of the structural complexity and diversity of glycan isomers, creating a need for analytical methods specifically suited to glycomics. Mass spectrometry, with its high sensitivity, high resolution, and high throughput, has become a central tool for glycomics analysis. Nevertheless, the intrinsically low ionization efficiency of glycans limits their detection sensitivity in complex biological samples. Furthermore, as glycomics data volumes increase, conventional spectral interpretation and database-comparison approaches no longer meet the demands of large-scale data processing. Thus, the development of efficient and accurate methods for glycan recognition and identification has become an active area of research.

To address these challenges, this work combines a chemical derivatization strategy with a liquid chromatography-mass spectrometry platform to improve the chromatographic separation performance and MS response of glycans. On this basis, we independently developed MS-GDF, a MATLAB-based software package for identifying derivatized glycans in high-resolution mass spectrometry data. It is used for systematic filtering and analysis of MS1 and MS/MS information for glycans.

MS-GDF uses a two-level identification strategy. **The first-level filter** uses the fixed mass difference between light- and heavy-isotope labeling reagents to pair precursor ions of derivatized glycans. **The second-level filter** employs a dedicated algorithm based on glycan structural characteristics: it calculates mass differences among MS/MS fragments and matches them against a monosaccharide-residue

database to select fragment-ion patterns consistent with glycan characteristics. The software is intended to address technical bottlenecks in glycan identification and enable comprehensive, rapid, and accurate analysis of glycans in complex samples. MS-GDF also has good scalability and may be extended to structurally analogous molecules formed by polymerization of a limited set of known monomeric units into complex long-chain structures, such as peptides and polymers. It has broad potential applications in disease diagnosis, therapeutic target discovery, identification of quality markers in foods and traditional Chinese medicines, and environmental pollutant monitoring.

This manual uses a reducing-end carbonyl-specific labeling strategy as an example. Human plasma N-glycans are derivatized with 2,4-bis(diethylamino)-6-hydrazinyl-1,3,5-triazine (T3) and its deuterated reagent, T3-D20, to illustrate the complete MS-GDF workflow.

1.2 Operating Environment

Hardware requirements: CPU clock frequency of 1.9 GHz or higher; hard-disk capacity of 500 GB or greater, with at least 4 GB of available space; and 8 GB RAM or greater.

Software requirements: 64-bit Windows 8/10/11 operating system, with MATLAB 2024a or later preinstalled, including the Bioinformatics Toolbox.

1.3 Software Installation

MS-GDF is a portable, installation-free application. It can be run directly after extraction, as shown in Figure 1.1. After importing data into the main interface, users can perform the corresponding data-processing tasks by following the operating steps.

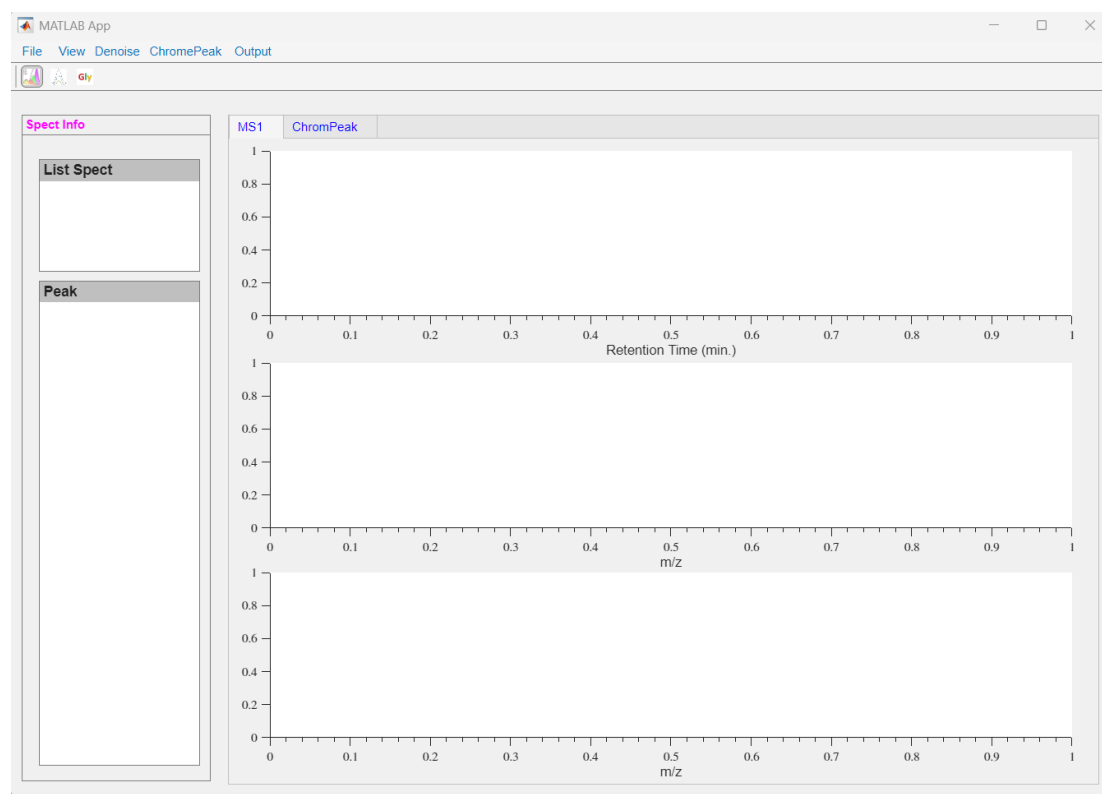


Figure 1.1 Main Interface of MS-GDF

1.4 Main Modules

The main modules of MS-GDF include data-file import and reading, parameter settings, spectrum visualization, and data-file export.

Chapter 2 Introduction to the MS-GDF Module Interface

The primary function of MS-GDF is to extract target analyte peak information using the filtering algorithms described above. The MS-GDF interface includes a menu bar, toolbar, data files, chromatographic result plots, mass spectral result plots, and extracted ions for chromatographic peaks, as shown in Figure 2.1.

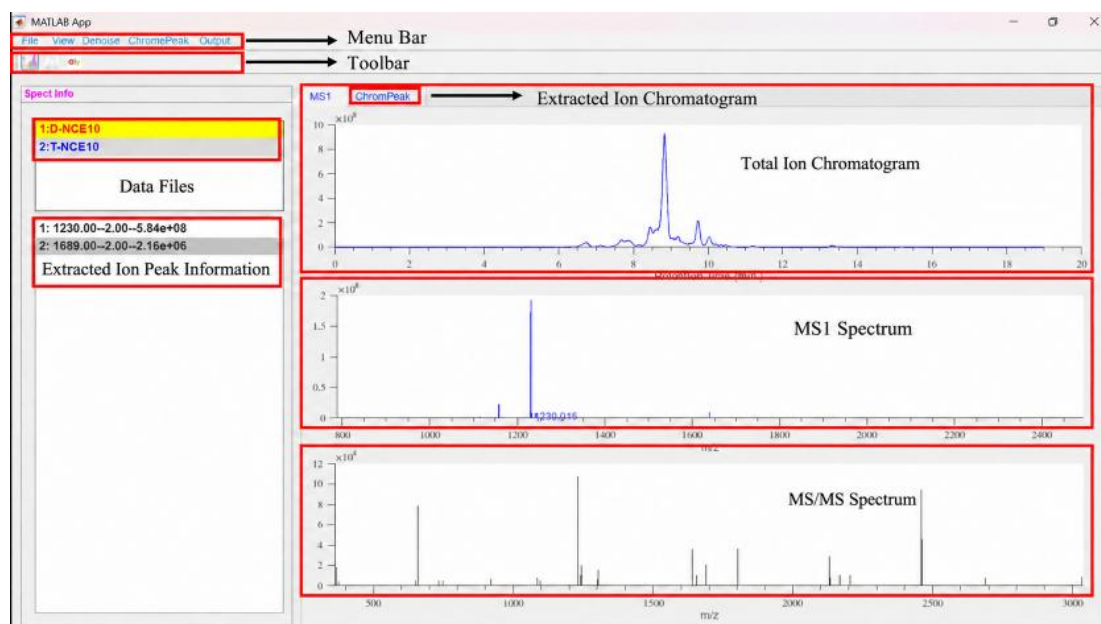


Figure 2.1 Main Module Interface After MS-GDF Is Launched

2.1 File Menu (File)

As shown in Figure 2.2, the File menu in the main interface includes file-format conversion, file import, label assignment, and file saving. The specific commands are described below.

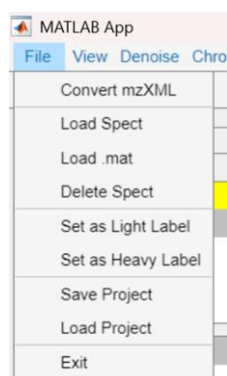


Figure 2.2 MS-GDF File Menu

【Convert mzXML】 Converts file data formats. Raw MS data in .mzXML format are converted to the MS-GDF-specific .lmss format, a lightweight MATLAB structured-data format.

【Load Spect】 Imports data files in .lmss format.

【Delete Spect】 Deletes data files in .lmss format.

【Set as Light Label】 Marks the selected file as light-labeled data. It is displayed as blue text on a gray background in the file list, as shown in Figure 2.3.

【Set as Heavy Label】 Marks the selected file as heavy-labeled data. It is displayed as red text on a yellow background in the file list, as shown in Figure 2.3.

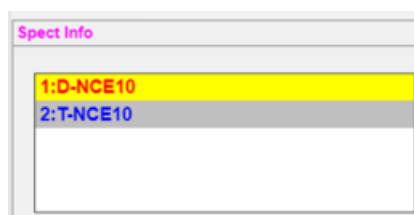


Figure 2.3 MS-GDF Data-File Labels

【Save Project】 Saves the project. The current file data can be saved at any stage of processing.

【Load Project】 Loads a previously saved project and its file data.

【Exit】 Exits and closes the MS-GDF main interface.

2.2 View Menu (View)

As shown in Figure 2.4, the View menu in the main interface is mainly used to set the displayed peak mass and retention time, as well as to undo and redo editing operations.

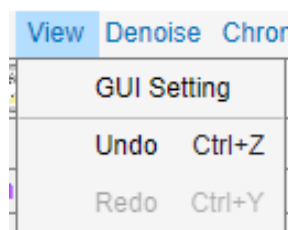


Figure 2.4 MS-GDF View Menu

【GUI Setting】 The parameter settings are shown in Figure 2.5. This

function sets the display ranges of chromatograms and mass spectra, including the mass-to-charge ratio (m/z) and retention-time (RT) ranges of target peaks.

“Axes limitation” : Sets axis limits.

Retention time region: Sets the retention-time range.

mass-to-charge region: Sets the mass-to-charge-ratio range.

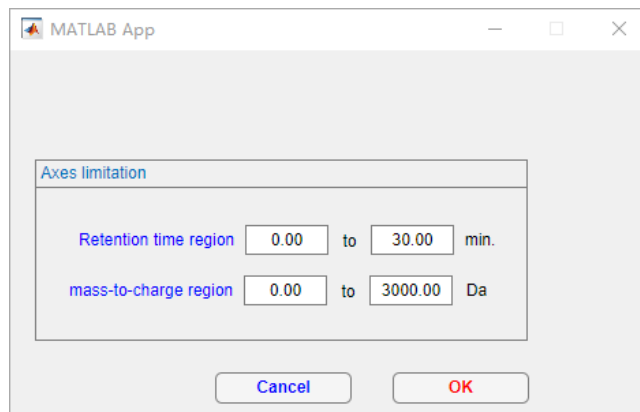


Figure 2.5 Denoise Parameter Settings Window

【Undo】 Returns to the previous operation (shortcut: Ctrl+Z).

【Redo】 Repeats the operation before the most recent undo (shortcut: Ctrl+Y).

2.3 Denoise

As shown in Figure 2.6, the principal function of the Denoise menu in the main interface is noise removal.

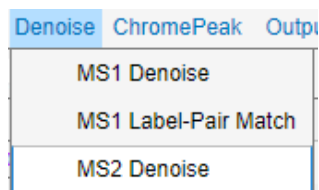


Figure 2.6 MS-GDF Data Denoising Menu

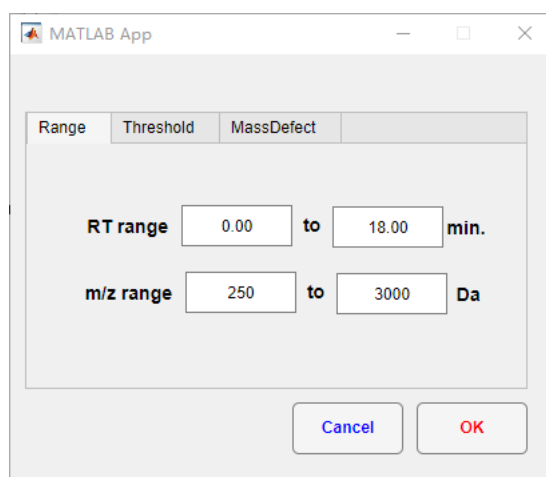
【MS1 Denoise】

The “Range” parameter settings are shown in Figure 2.7.

RT range: Specifies the retention-time window to be processed; only mass spectral data within this range are analyzed.

m/z range: Specifies the mass-to-charge-ratio range to be processed; only mass

spectral data within this range are analyzed.



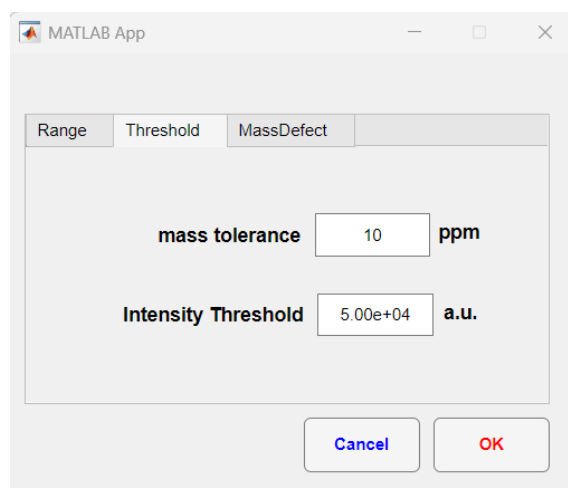
A screenshot of a MATLAB App window titled "MATLAB App". It features a tabbed interface with three tabs: "Range", "Threshold", and "MassDefect". The "Range" tab is currently selected. Inside the tab, there are two rows of input fields. The first row is labeled "RT range" and contains two text boxes with values "0.00" and "18.00", followed by the text "min.". The second row is labeled "m/z range" and contains two text boxes with values "250" and "3000", followed by the text "Da". At the bottom right of the window, there are two buttons: "Cancel" and "OK".

Figure 2.7 Range Parameter Settings Window

The “Threshold” parameter settings are shown in Figure 2.8.

mass tolerance: Sets the m/z matching tolerance.

Intensity Threshold: Sets the minimum ion signal intensity.



A screenshot of a MATLAB App window titled "MATLAB App". It features a tabbed interface with three tabs: "Range", "Threshold", and "MassDefect". The "Threshold" tab is currently selected. Inside the tab, there are two rows of input fields. The first row is labeled "mass tolerance" and contains a text box with the value "10", followed by the text "ppm". The second row is labeled "Intensity Threshold" and contains a text box with the value "5.00e+04", followed by the text "a.u.". At the bottom right of the window, there are two buttons: "Cancel" and "OK".

Figure 2.8 Threshold Parameter Settings Window

The “Mass Defect” setting specifies mass-defect ranges for non-target peaks, as shown in Figure 2.9. This function is primarily used to remove background interference ions with characteristic mass defects and is generally recommended to be disabled for glycan analysis.

from: Minimum mass-defect value for non-target peaks.

to: Maximum mass-defect value for non-target peaks.

Add: Adds the mass-defect range for non-target peaks specified above to the list

on the right.

Del: Deletes the selected mass-defect range for non-target peaks from the list on the right.

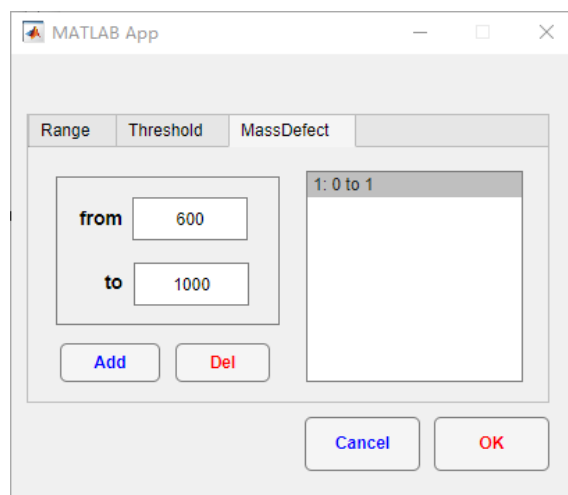


Figure 2.9 Mass Defect Parameter Settings Window

【MS1 Label-Pair Match】 Performs strong denoising of all non-target MS points based on isotope MS-point matching rules. The parameter settings are shown in Figure 2.10.

m/z Tolerance: Sets the allowable mass deviation for light/heavy-label pairing. It should be set according to mass-spectrometer resolution; 5-10 ppm is recommended for high-resolution mass spectrometry.

RT Tolerance: Sets the allowable retention-time difference for matching light- and heavy-labeled data.

Label m/z: Mass difference between the light and heavy isotope labels.

#Charge: Maximum charge state of target analytes. T3-derivatized N-glycans commonly carry +1 to +3 charges in an electrospray ion source; a value of 3 is recommended.

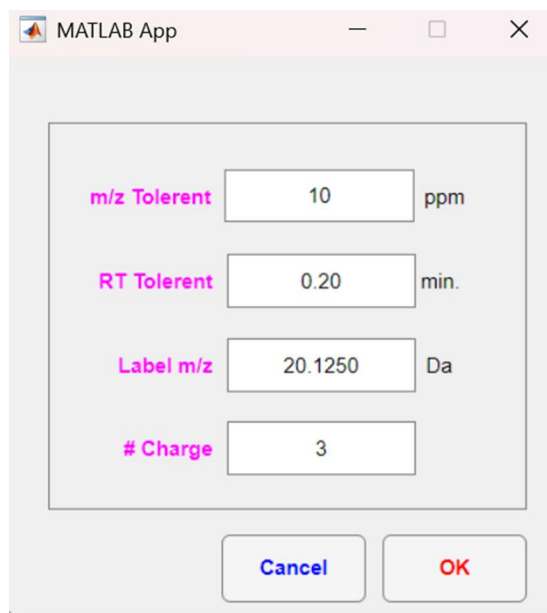


Figure 2.10 MS1 Label-Pair Match Parameter Settings Window

【MS2 Denoise】 Denoises MS/MS data. The parameter settings are shown in Figure 2.11.

Keep top-K m/z peaks: Retains the top K fragment-ion peaks by abundance.

minimal intensity: Sets the minimum fragment-ion signal intensity.

minimal m/z: Sets the minimum fragment-ion mass-to-charge ratio.

maximal m/z: Sets the maximum fragment-ion mass-to-charge ratio.

m/z Tolerent: Sets the error range for fragment-ion mass-to-charge ratios.

Max Charges: Maximum charge number of fragment ions.

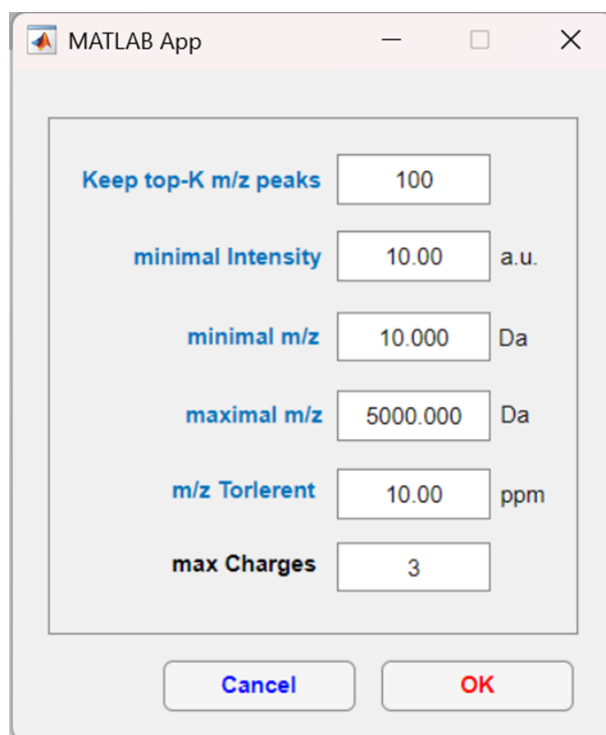


Figure 2.11 MS2 Denoise Parameter Settings Window

2.4 Chromatographic Peak Menu (ChromePeak)

The chromatographic peak menu primarily includes chromatographic peak extraction, scoring and ranking of chromatographic peaks according to peak shape, light/heavy-label matching, oligosaccharide-residue matching, and fragment-ion peak matching, as shown in Figure 2.12.

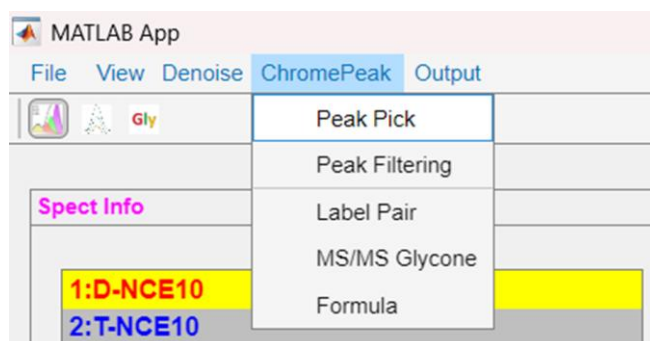


Figure 2.12 Peak Matching Menu Items

【Peak Pick】 Peak-extraction function that extracts all valid peaks, as shown in Figure 2.13.

“ROI screen”

m/z tolerance: Mass-to-charge-ratio tolerance for chromatographic peak extraction. The m/z deviation between adjacent scans within the same peak must be less than this value.

min. scans: Sets the minimum number of scans for peak screening.

max consecutive missing: Sets the maximum number of missing scans tolerated within a single peak trace.

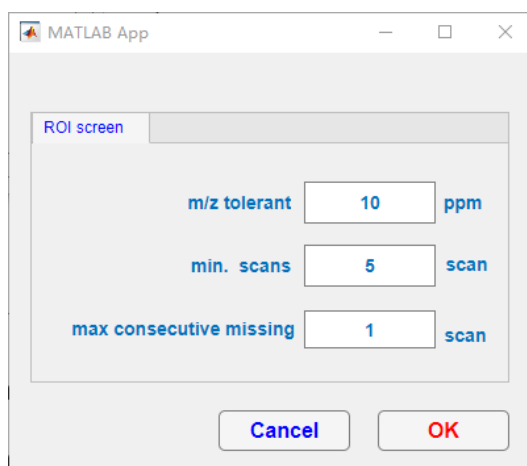


Figure 2.13 Peak-Extraction Parameter Settings

【Peak Filtering】 Removes isotope peaks based on ^{13}C isotope patterns. Because glycans have high molecular masses, ^{13}C isotope peaks can have substantial abundance. This step identifies and removes isotope peaks according to the theoretical isotope distribution, including the fixed $^{13}\text{C}/^{12}\text{C}$ mass difference and abundance ratio at different charge states, retaining only the monoisotopic peak, as shown in Figure 2.14.

Isotopic mass: Mass difference between isotope peaks, typically ^{13}C isotope peaks.

RT tolerance: Retention-time difference.

m/z tolerance: Mass-to-charge-ratio error range.

max Charges: Maximum charge setting.

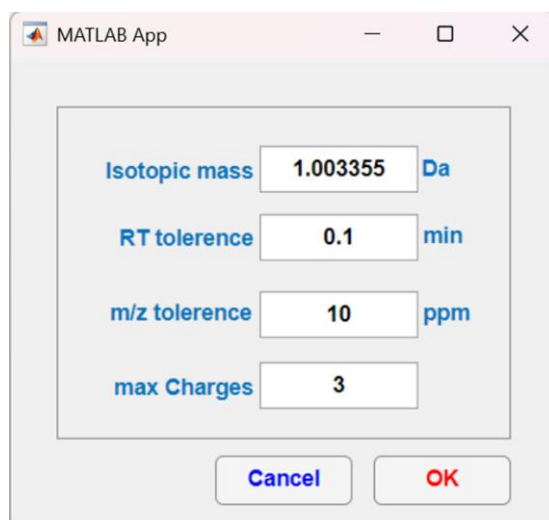


Figure 2.14 Isotope-Peak Filtering Parameter Settings

【Label Pair】 Performs peak matching among chromatographic peaks based on the fixed mass difference between light and heavy isotopes, as shown in Figure 2.15.

m/z tolerant: Sets the m/z error tolerance for the fixed mass difference between light- and heavy-isotope labels.

RT tolerant: Sets the maximum allowable retention-time difference between light- and heavy-isotope peaks.

Isotope: Fixed mass difference generated by the two derivatization products.

Charge: Maximum charge number of ion peaks.

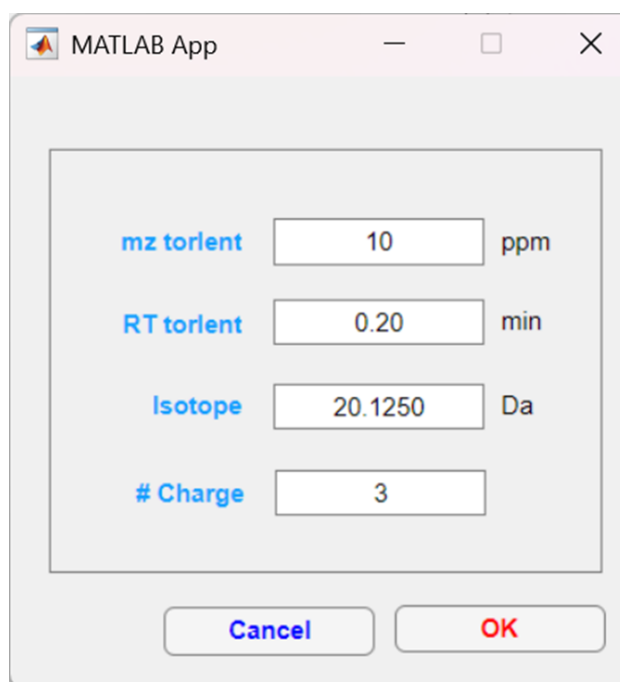


Figure 2.15 Light/Heavy Isotope-Labeled Peak-Matching Parameter Settings

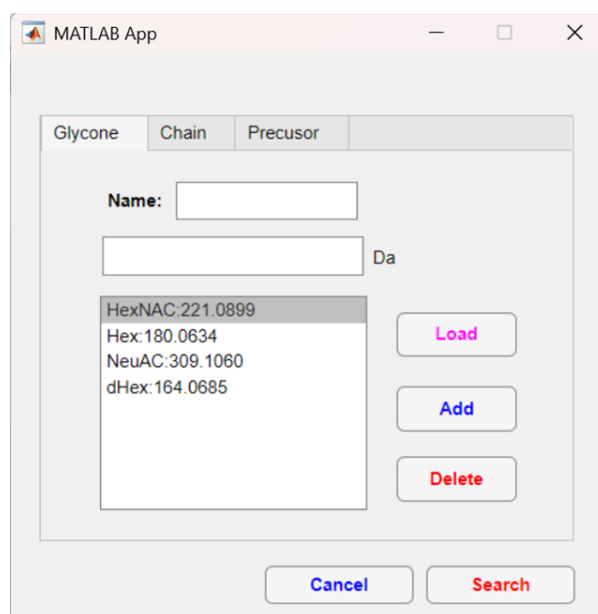
【MS/MS Glycone】 Residue-matching menu.

“Glycone” : Enter the names of monosaccharide residues, such as Hex, HexNAc, and NeuAc, and their exact monoisotopic masses, as shown in Figure 2.16.

Load: Imports a list of monosaccharide residue masses in .xlsx format.

Add: Manually enters monosaccharide residue masses.

Delete: Deletes the selected monosaccharide residue masses.



The screenshot shows a MATLAB App window titled 'MATLAB App'. Inside, there's a tabbed interface with 'Glycone', 'Chain', and 'Precursor' tabs. The 'Glycone' tab is active. It contains a 'Name:' label followed by a text input field. Below this is another text input field followed by 'Da'. A list box contains the following text: 'HexNAc:221.0899', 'Hex:180.0634', 'NeuAc:309.1060', and 'dHex:164.0685'. To the right of the list box are three buttons: 'Load' (pink), 'Add' (blue), and 'Delete' (red). At the bottom of the window are two buttons: 'Cancel' (blue) and 'Search' (red).

Figure 2.16 Input of Monosaccharide Residue Masses

“Chain” : Constructs an oligosaccharide residue mass pool, as shown in Figure 2.17.

m/z of Condense Group: Set to 18.01050, indicating that two monosaccharides are linked through loss of one water molecule. Enter the exact molecular mass of H₂O in this field.

Glycone Combination: Sets the maximum number of monosaccharide residues in a residue combination. For example, when set to 4, the mass pool includes all possible combinations of 1-4 monosaccharide residues, with dehydration, corresponding to Y1-Y4 or B1-B4 fragment ions.

Combine: Adds residue masses according to the previous settings; the generated oligosaccharide residue mass pool is displayed in the right-hand panel.

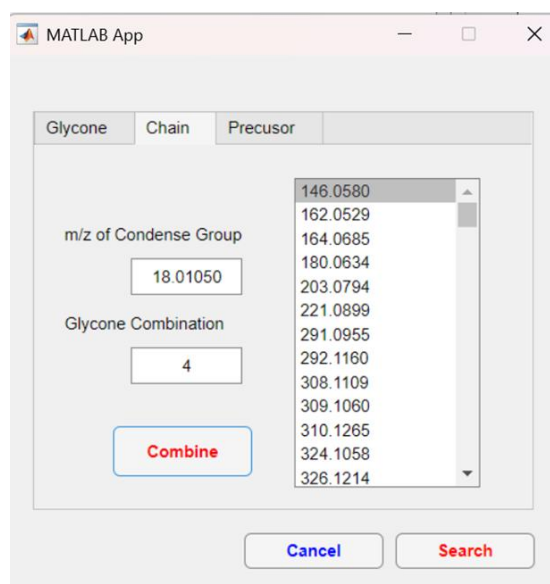


Figure 2.17 Construction of the Oligosaccharide Residue Mass Pool

“Precursor” : Pairwise calculates mass differences among the top K fragment ions by abundance in MS2 spectra and matches the differences to the oligosaccharide residue mass pool, as shown in Figure 2.18.

Precursor Window: Sets the m/z matching tolerance between the precursor ion of an MS/MS spectrum and the peak matched at the MS1 level.

m/z Tolerance: Sets the m/z error range for matching.

minimal chain: Minimum number of matches between fragment-ion mass differences and entries in the mass pool.

RT Tolerance: Sets the allowable retention-time difference between the precursor ion for which MS/MS fragments were acquired and the chromatographic peak matched as a light/heavy-label pair.

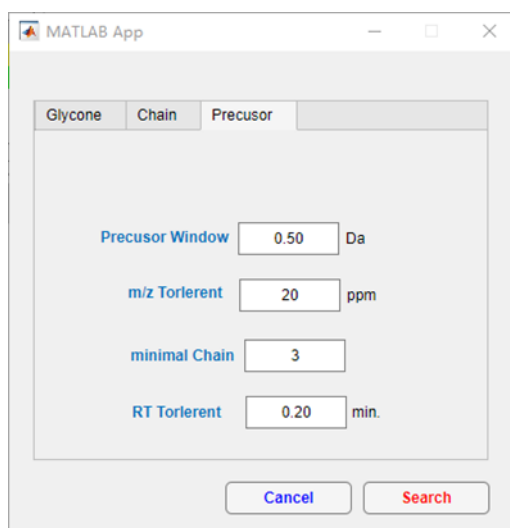


Figure 2.18 Matching of Pairwise Mass Differences in MS/MS Data to the Mass Pool

【Formula】 Calculates possible monosaccharide compositions for potential glycans identified in the preceding steps, as shown in Figure 2.19.

Glycone List: Possible monosaccharide compositions and their masses. These are determined by the monosaccharides entered when constructing the oligosaccharide residue mass pool in the preceding “Chain” step and cannot be modified here.

Adduct Ion List: Possible adduct-ion types and their masses, such as H^+ , Na^+ , K^+ , and NH_4^+ .

T3: Actual mass of the derivatization adduct.

m/z tol.: Allowable mass deviation for molecular formula calculation.

Load: Imports possible ion-adduct types from an Excel spreadsheet.

Delete: Deletes the selected ion-adduct type.

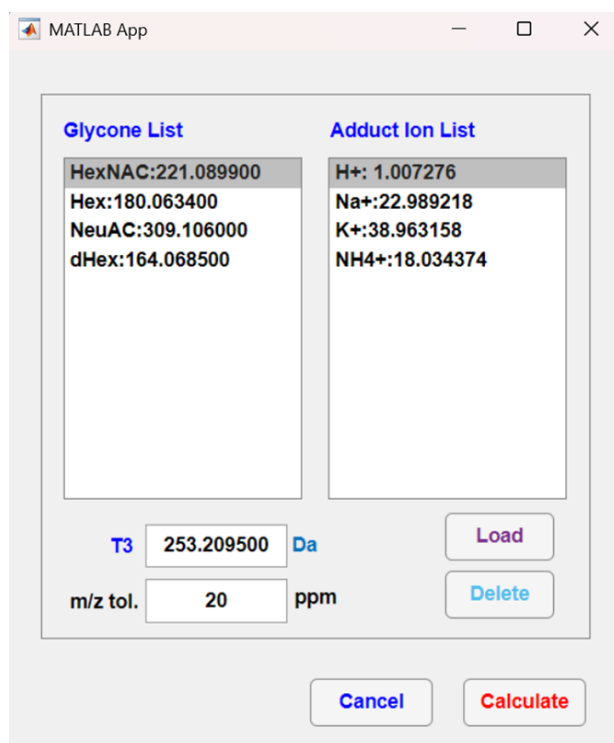


Figure 2.19 Calculation of Possible Molecular Formula Compositions

2.5 Results Output Menu (Output)

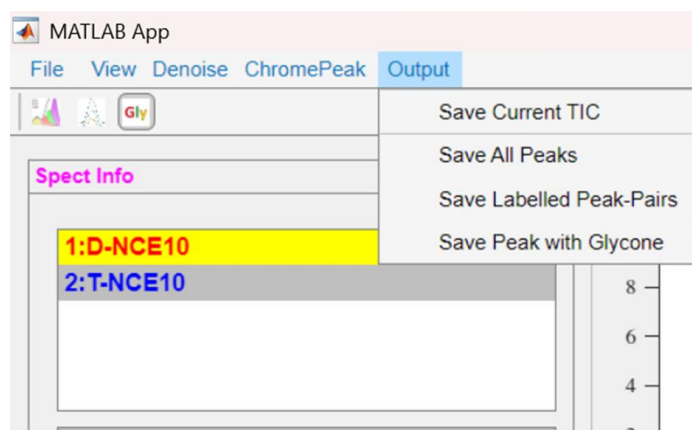


Figure 2.20 Results Output Menu

【Save Current TIC】 Exports the total ion chromatogram (TIC) or mass spectrum displayed in the current view as an image file.

【Save All Peaks】 Exports mass spectral peaks after 【Peak Pick】 as a list in .xlsx format.

【Save Labelled Peak-Pairs】 Exports target peaks after the 【Label Pair】 operation.

【Save Peak with Glycone】 Exports target peaks after the **【MS/MS Glycone】** operation.

Chapter 3 Example Workflow for MS-GDF Software

T3 and T3-D20 are used to derivatize human plasma separately before mass spectrometric acquisition. The resulting mass spectrometry data files are used to demonstrate the software workflow. The LC-MS data are named T-NCE10 and D-NCE10, respectively.

3.1 Format Conversion

First, use the format-conversion software “MSConvert” to convert the acquired raw data from “Raw” format to “.mzXML” format. Then open the main interface and select File → Convert mzXML. In the dialog that opens, click Add files to select the .mzXML files to be converted; batch selection is supported. After confirmation, click Convert to begin file-format conversion. When conversion is complete, a “Conversion Completed!” dialog appears, as shown in Figure 3.1, and the converted files are saved in the folder containing the original data.

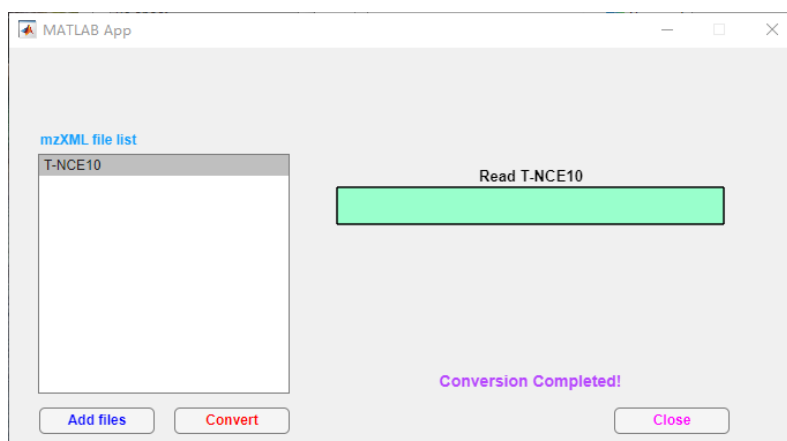


Figure 3.1 Format Conversion Dialog

3.2 Data Import

Select File → Load Spect to import, in sequence, the converted light- and heavy-isotope derivatization data files, “T-NCE10.lmss” and “D-NCE10.lmss”. Files in .lmss format contain the complete mass spectrometry information for the sample, including MS1 full-scan data and MS/MS data acquired in data-dependent acquisition

(DDA) mode. The imported data are shown in Figure 3.2.

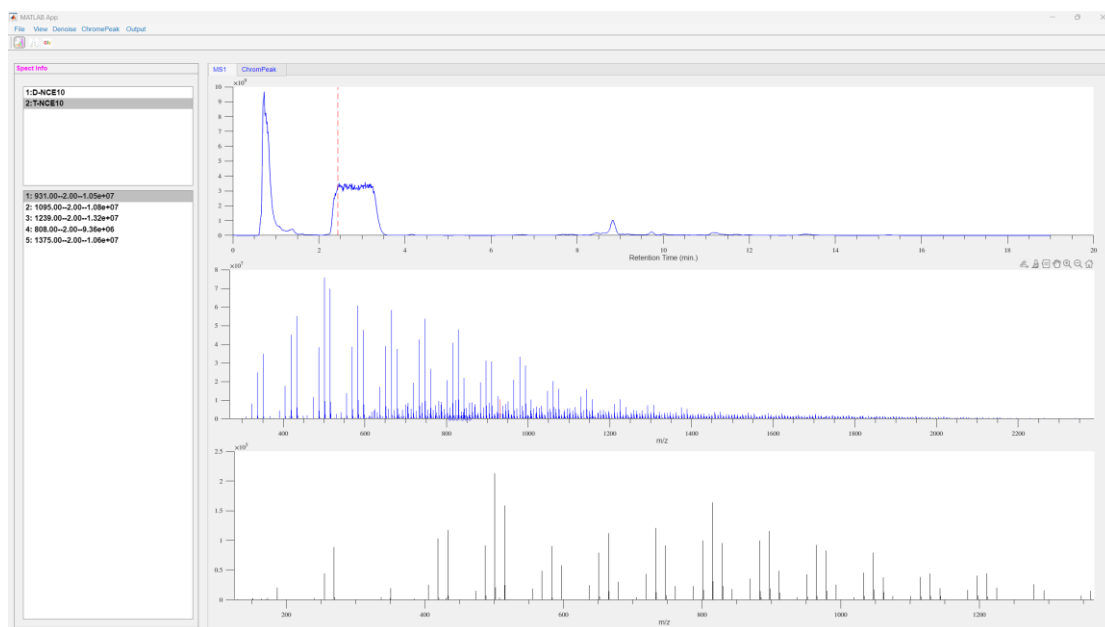


Figure 3.2 Main Software Interface After Data Import

Label the imported data by selecting 1.T-NCE10 and choosing File → Set as Light Label, then selecting “2.D-NCE10” and choosing File → Set as Heavy Label.

3.3 Denoising

Select Denoise → MS1 Denoise to open the parameter-setting dialog. Set parameters in each field according to the conditions of this example, including the retention-time range and intensity range of the target analytes, as illustrated in Figures 3.3-3.5.

Based on the chromatographic behavior of N-glycans, set the retention-time window to 4-18 min. An m/z range of 500-2500 Da covers most multiply charged T3-derivatized N-glycans.

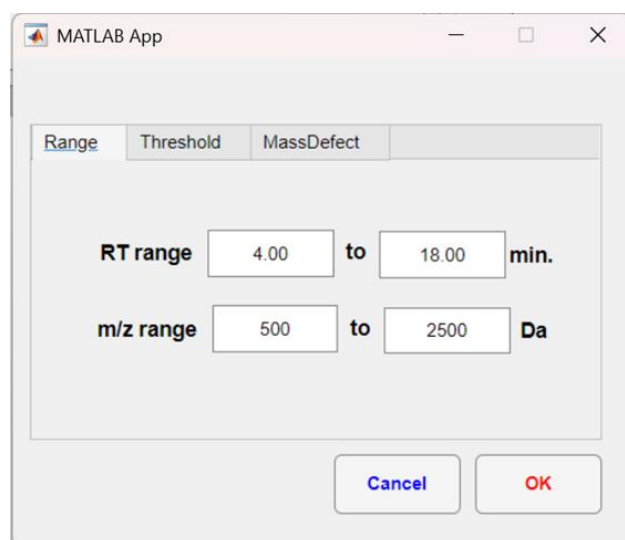
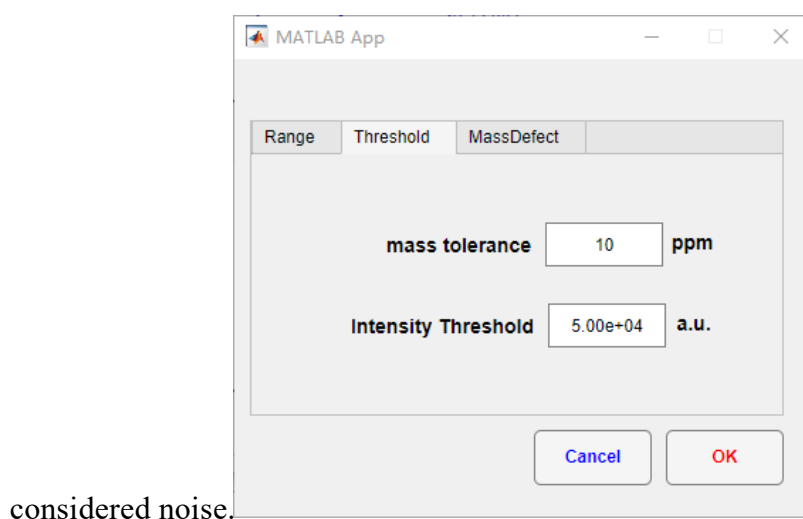


Figure 3.3 Denoise-Range Parameter Settings Window

Set the intensity threshold to 5.00×10^4 , or 50000, a.u. Signals below this value are



considered noise.

Figure 3.4 Denoise-Threshold Parameter Settings Window

In this example, no mass-defect restriction is applied because glycans exhibit a broad mass-defect range.

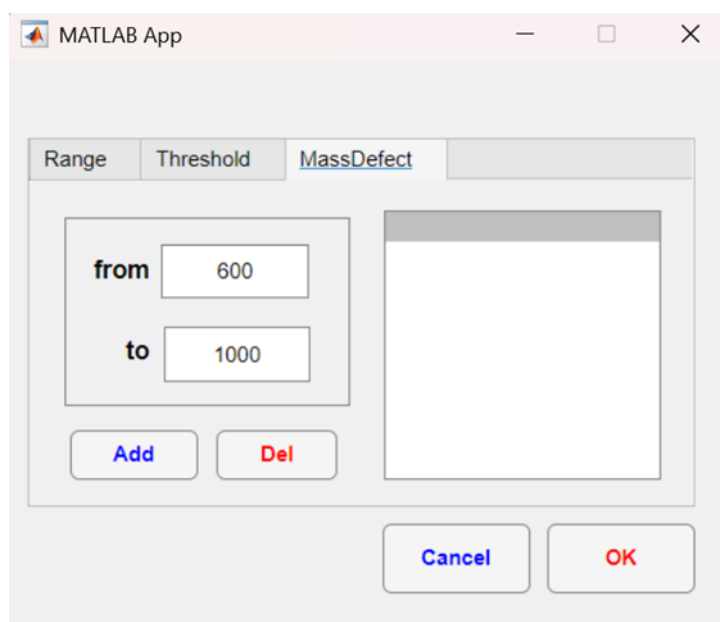


Figure 3.5 Denoise-Mass Defect Parameter Settings Window

Select Denoise → MS1 Label-Pair Match to open the parameter-setting dialog. Set parameters according to the conditions of this example. Here, the mass difference between T3 and T3-D20 is 20.1250, and derivatives of the same glycan have highly similar retention times; therefore, the retention-time tolerance is set to 0.2 min.

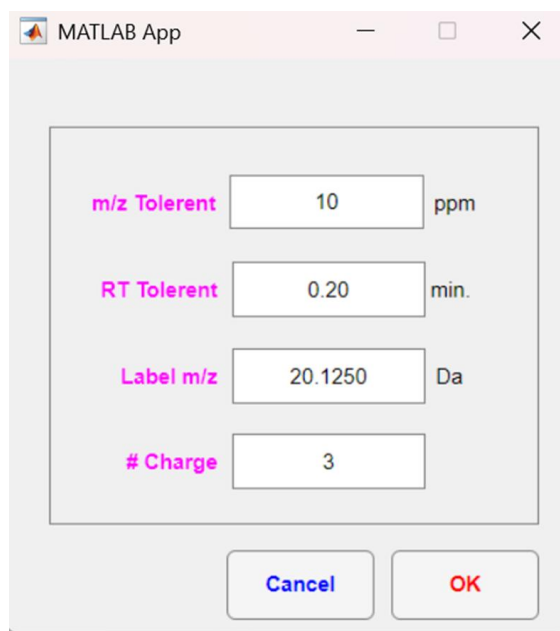


Figure 3.6 Denoise-Range Parameter Settings Window

Select Denoise → MS2 Denoise to open the parameter-setting dialog. For this example, retain the 20 most abundant fragment ions; set the minimum m/z to 500, the

maximum m/z to 5000, the tolerance to 10 ppm, and the maximum charge number to 3, as shown in Figure 3.7.

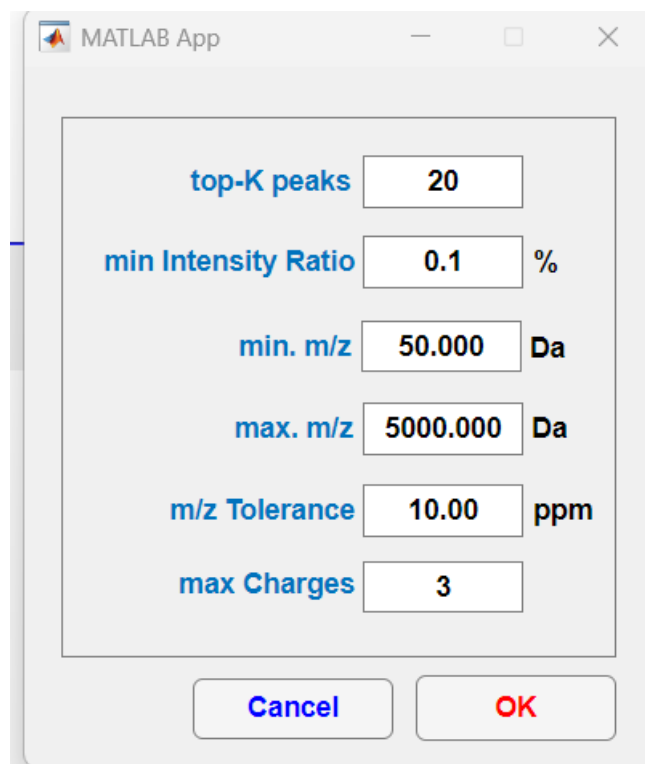


Figure 3.7 Denoise-Range Parameter Settings Window

Once all Denoise menu operations are complete, the interface is as shown in Figure 3.8, and the data have been substantially simplified by denoising.

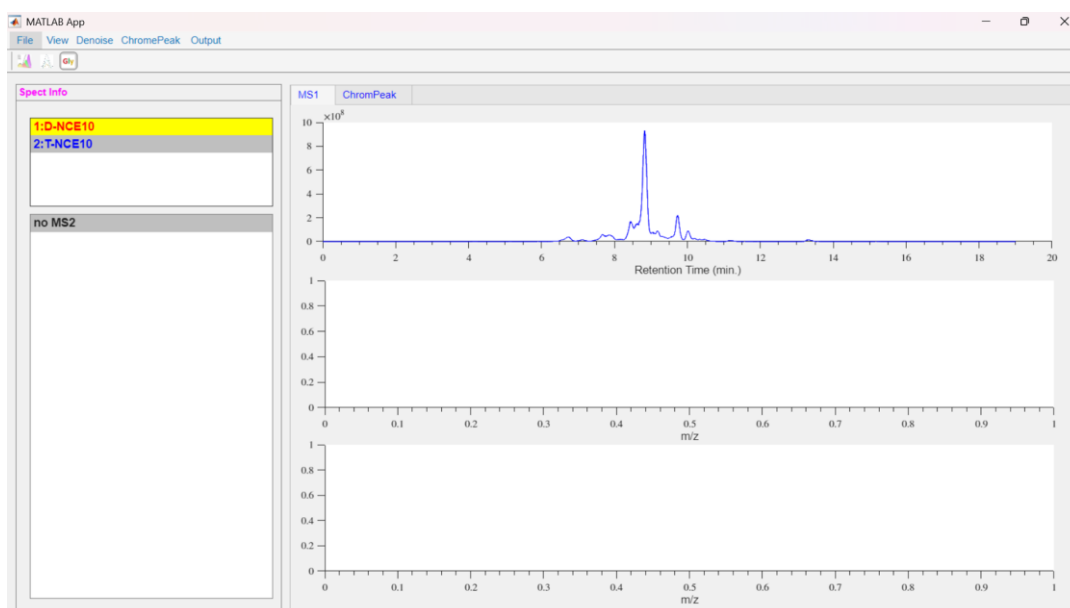


Figure 3.8 Main Software Interface After Denoise Processing

3.4 Chromatographic Peak Identification

Select ChromePeak → Peak Pick to open the parameter-setting dialog. Set parameters according to the conditions of this example, as shown in Figure 3.9. Here, an m/z tolerance of 10 ppm is used, as recommended for high-resolution mass spectrometry; the minimum number of consecutive scans is set to 10, corresponding to approximately 0.1-0.2 min depending on the acquisition frequency; and the relative intensity threshold is set to 1% to remove shoulder peaks and baseline noise.

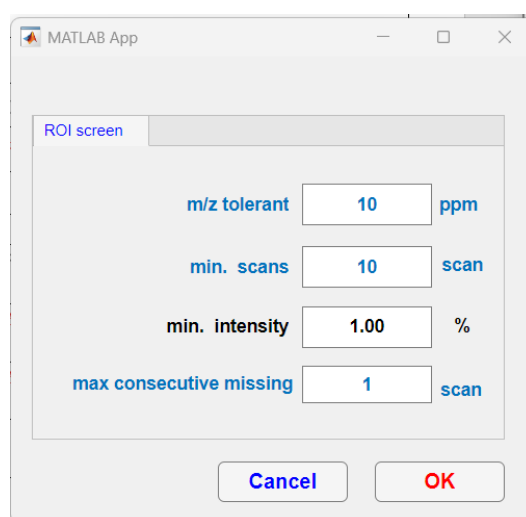


Figure 3.9 Peak Identification Parameter Settings

After processing, the interface shown in Figure 3.10 lists all chromatographic peak information extracted from the two data files. In this example, 1122 and 1556 chromatographic peaks were identified in the light- and heavy-labeled files, respectively.

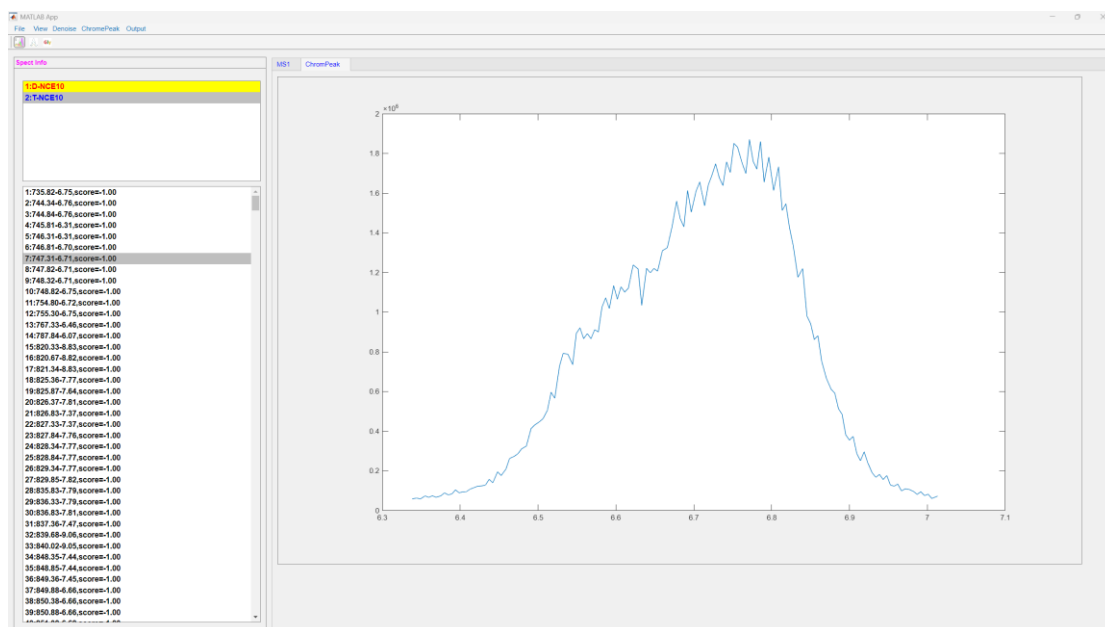


Figure 3.10 Main Software Interface After Peak Identification

3.5 Isotope-Peak Filtering

Select ChromePeak → Peak Filtering to open the parameter-setting dialog. Choose removal of isotope peaks caused by ^{13}C ; the parameter settings are shown in Figure 3.11.

Figure 3.11 Isotope-Peak Filtering Parameter Settings

After processing, 465 peaks remained in the light-labeled data and 676 peaks

remained in the heavy-labeled data. This indicates that more than 50% of the chromatographic peaks obtained in the preceding peak-extraction step were isotope peaks and were removed in this step. The results are shown in Figure 3.12.

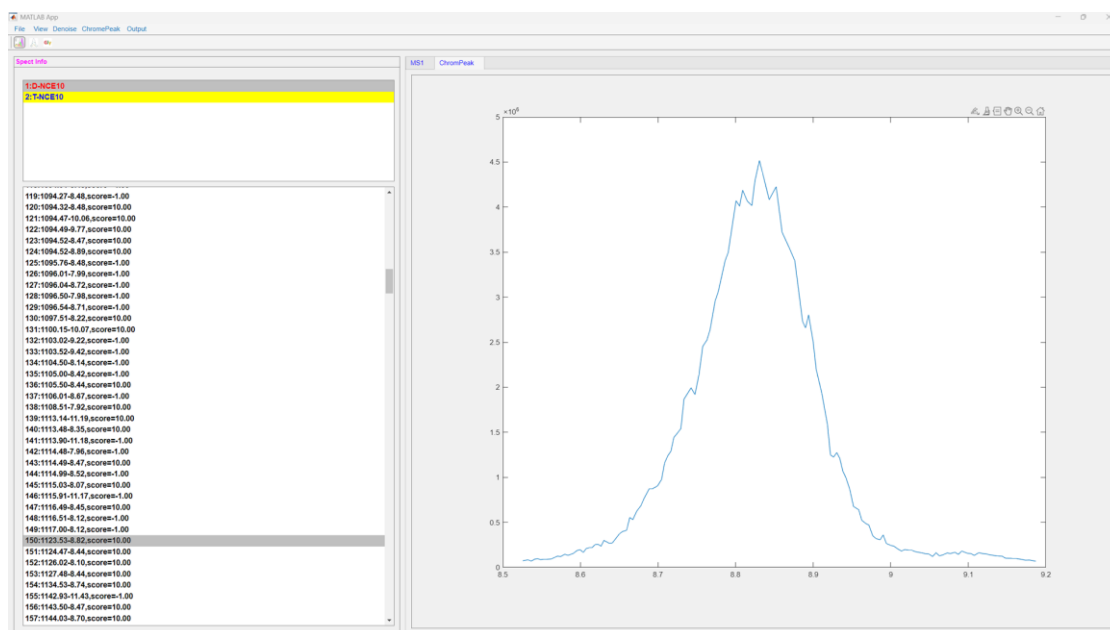


Figure 3.12 Software Interface After Isotope Filtering

3.6 Light/Heavy Isotope-Labeled Peak Matching

Select ChromePeak → Label Pair to open the parameter-setting dialog. Set parameters according to the conditions of this example, as shown in Figure 3.13.

The isotope derivatization reagents used in this example are T3 and T3-D20. In T3-D20, 20 hydrogen atoms of T3 are replaced by deuterium, resulting in a fixed mass difference of 20.1250 Da. For T3-derivatized N-glycans, charges of +1 to +3 are commonly observed; therefore, set “# Charge” to 3. After running the software, light/heavy-label paired peaks are displayed in the left panel, as shown in Figure 3.14. Under these parameter settings, 302 chromatographic peaks representing potential glycan components were identified.

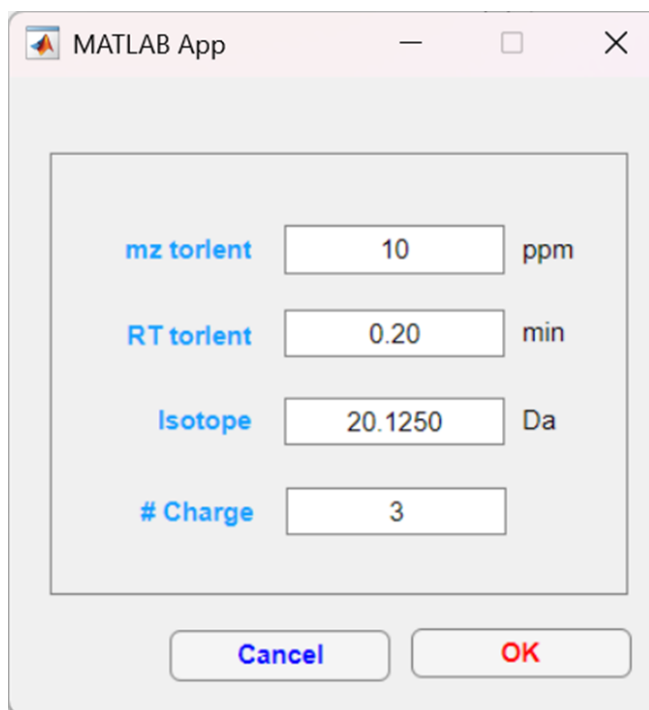


Figure 3.13 Light/Heavy Isotope-Labeled Peak-Matching Parameter Settings

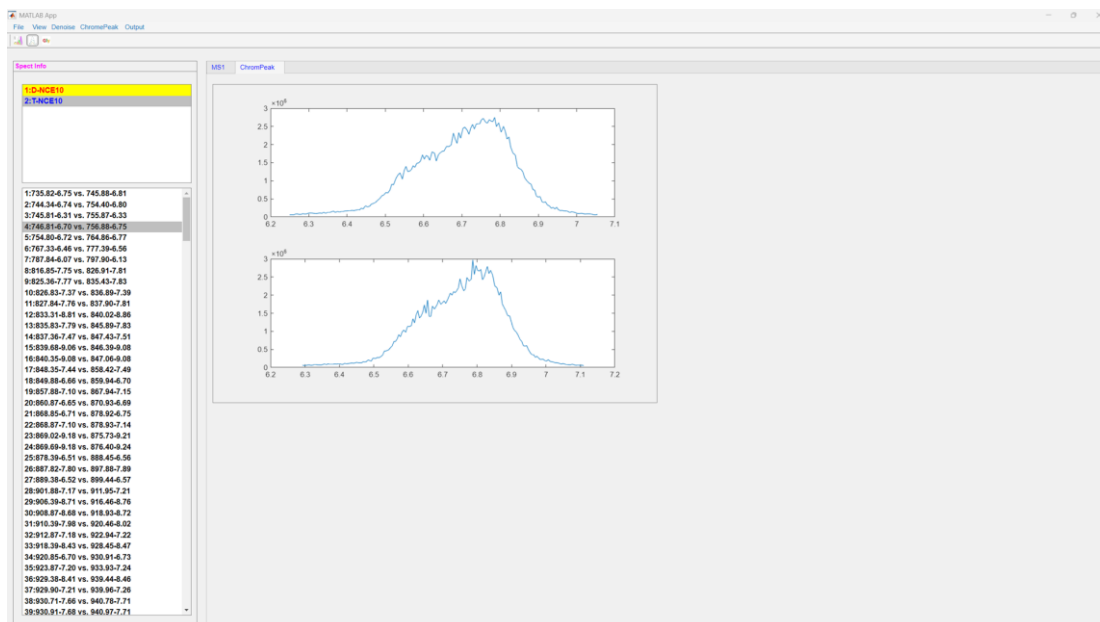


Figure 3.14 Software Peak Pairs Interface After Peak Matching

3.7 Residue Matching

Select ChromePeak → MS/MS Glycone to open the parameter-setting dialog. Set parameters according to the conditions of this example.

In the “Glycone” panel, click Load to import a .xlsx list of oligosaccharide residue names and masses. Alternatively, use Add to enter their names and masses

manually, as shown in Figure 3.15.

The MATLAB App window displays the 'Glycone' panel. It includes a 'Name' input field, a mass input field labeled 'Da', and a list of glycone masses: HexNAC:221.0899, Hex:180.0634, NeuAC:309.1060, and dHex:164.0685. To the right of the list are buttons for 'Load' (pink), 'Add' (blue), and 'Delete' (red). At the bottom of the panel are 'Cancel' (blue) and 'Search' (red) buttons.

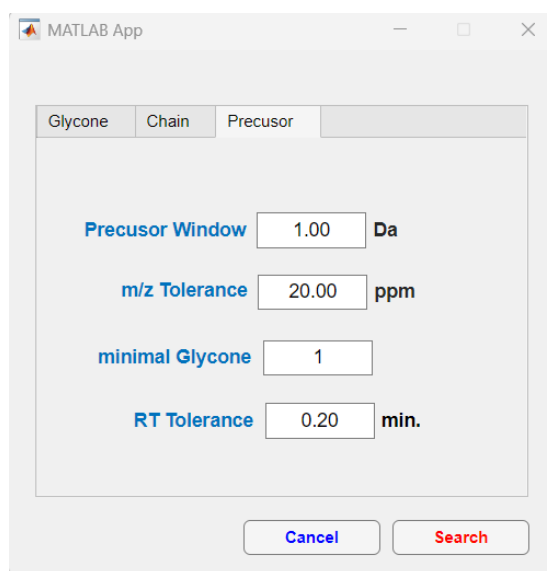
Figure 3.15 Input of Oligosaccharide Residue Masses

Click the “Chain” panel. In this example, two monosaccharides combine into a disaccharide through loss of one water molecule; therefore, enter 18.0105 for “m/z of Condense Group”. Then enter 3 for “Glycone Combination” and click Combine. The monosaccharides entered in the preceding step are combined as dimers and trimers to form a mass pool of all possible MS/MS residues, as shown in Figure 3.16.

The MATLAB App window displays the 'Chain' panel. It includes an 'm/z of Condense Group' input field with the value 18.01050, a 'Glycone Combination' input field with the value 4, and a 'Combine' button (red). To the right is a list of resulting mass values: 146.0580, 162.0529, 164.0685, 180.0634, 203.0794, 221.0899, 291.0955, 292.1160, 308.1109, 309.1060, 310.1265, 324.1058, and 326.1214. At the bottom of the panel are 'Cancel' (blue) and 'Search' (red) buttons.

Figure 3.16 Construction of the Oligosaccharide Residue Mass Pool

Click the “Precursor” panel and set parameters according to the conditions of this example, as shown in Figure 3.17. Using the selected settings, the software pairwise calculates mass differences between the m/z values of fragment ions produced by acquired precursor ions and matches these differences to mass values in the mass pool. A precursor ion is retained when the number of successfully matched fragment-ion mass differences is at least 1.



The image shows a MATLAB App window titled 'MATLAB App'. It has three tabs: 'Glycone', 'Chain', and 'Precursor'. The 'Precursor' tab is selected. Inside the tab, there are four input fields with labels in blue text: 'Precursor Window' with a value of '1.00' and unit 'Da'; 'm/z Tolerance' with a value of '20.00' and unit 'ppm'; 'minimal Glycone' with a value of '1'; and 'RT Tolerance' with a value of '0.20' and unit 'min.'. At the bottom of the window, there are two buttons: 'Cancel' and 'Search'.

Figure 3.17 Matching of Pairwise Mass Differences in MS/MS Data to the Mass Pool

After processing, the results interface is shown in Figure 3.18. Under the parameter settings in this example, 102 potential glycan components were found, and their precursor ions and retention times are displayed in the left panel.

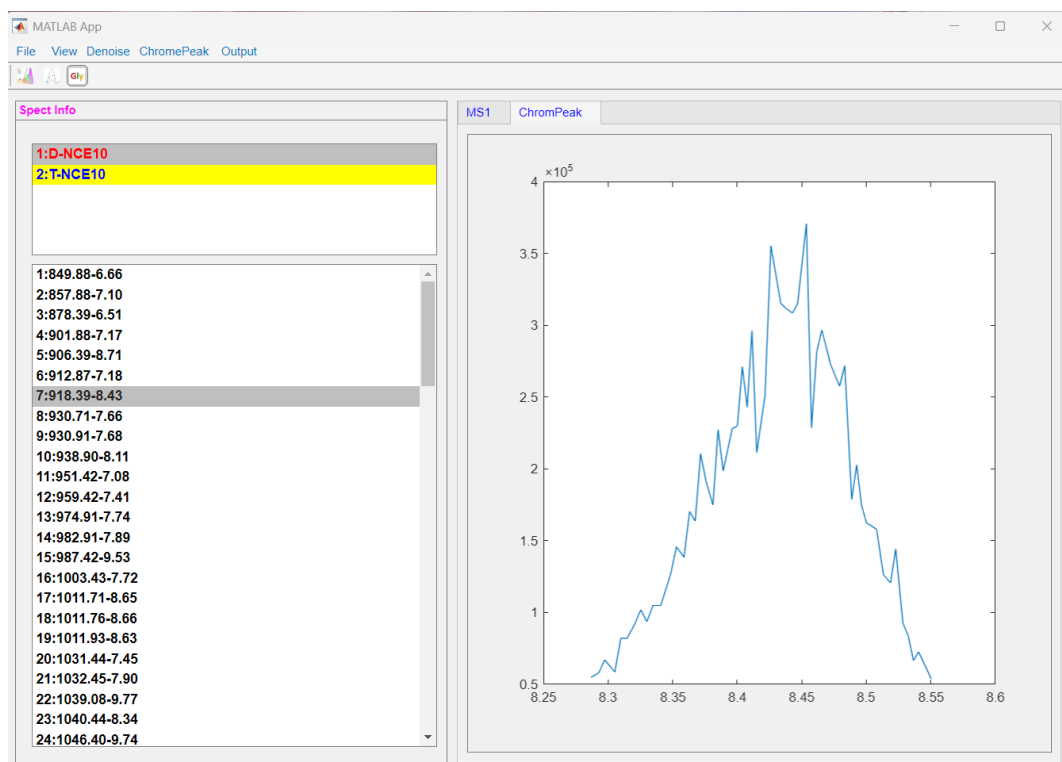


Figure 3.18 Software Interface After MS/MS Residue Matching

3.8 Molecular Formula Calculation

Select ChromePeak → formula to open the dialog shown below. The Glycone List on the left contains the monosaccharide compositions entered in the preceding steps. The Adduct Ion List on the right contains possible adduct-ion types, which can be uploaded through Load. T3 is the derivatization reagent used in this study and can be modified according to the actual experimental conditions. The parameter settings are shown in Figure 3.19.

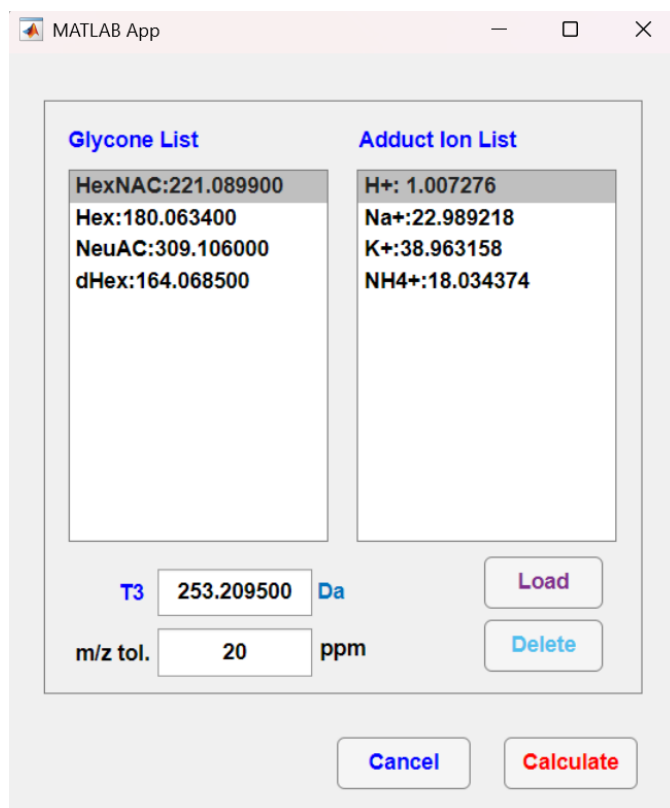


Figure 3.19 Molecular Formula Calculation Parameter Settings

3.9 Results Export

When data processing is complete, the selected target data peaks can be exported as tables, and the processed chromatograms can be exported as images.

Export as an image:

Select Output → Save Current TIC to export the TIC chromatogram or mass spectrum displayed in the current window.

Export as a list:

Select Output → Save All Peaks to open the file-save dialog and save the file in .xlsx format. The exported data consist of mass spectral peak information identified after **【Peak Pick】**, including the retention time, m/z, ion intensity, and charge state of T-NCE10. The results are shown in Figure 3.20.

RT (min)	m/z	Intensity	#Charge
6.74644365	735.822303	88826.9844	2
6.74027879	744.335865	117046.883	2
6.30859302	745.805011	282385.469	2
6.70214952	746.812901	2754944	2
6.72088006	754.798343	167984.984	2
6.45596968	767.326142	97215.6406	2
6.06776599	787.839715	84676.5469	2
7.75187566	816.849558	68244.5625	2
7.77455721	825.362389	231382.141	2
7.36667461	826.831843	231810.016	2
7.75598158	827.83974	1768703.38	2
8.80636074	833.309868	97330.4531	3
7.78892825	835.825468	205203.219	2
7.46821445	837.361946	78385.1563	2
9.05557603	839.684285	83883.6797	3
9.08098577	840.347378	64054.7344	3
7.43847861	848.353011	386537.344	2
6.66362449	849.878127	2107797.25	2
7.1012325	857.875487	477515.375	2

Figure 3.20 List Exported by 【Save All Peaks】

Select Output → Save Labelled Peak-Pairs to open the file-save dialog and save the file in .xlsx format. The exported data consist of target peak information retained after 【Label Pair】, including the retention times, m/z values, ion intensities, and charge states of paired T-NCE10 and D-NCE10 precursor ions. The results are shown in Figure 3.21.

RT: Retention time; RT-1 denotes the light-label retention time, and RT-2 denotes the heavy-label retention time.

m/z: Mass-to-charge ratio; m/z-1 denotes the light-label m/z, and m/z-2 denotes the heavy-label m/z.

intensity: Peak intensity at the current retention time.

#charge: Charge state.

Score: Charge-state scoring system; confidence in charge-state determination based on isotope-pattern recognition. Score 10: the charge states of both light- and heavy-labeled ions are automatically confirmed by ¹³C isotope distributions; Score 5: only the light-labeled ion is confirmed; Score 3: only the heavy-labeled ion is confirmed; Score 0: neither ion is confirmed.

#Glycone: Number of residue types matched by fragment ions.

Similarity: Chromatographic peak similarity between the light- and heavy-labeled ions. Higher similarity indicates a more plausible pair.

Formula: Molecular formula calculated from the m/z of the light-labeled ion.

Glycone#n: Glycan types in the mass pool matched by residues.

RT-1 (m/z-1)	Intensity-1	RT-2 (m/z-2)	Intensity-2	#Charge	Score	#Glycone	Similarity	Formula#1	Formula#2	Glycone#1	Glycone#2
8.81	1230.00025	219205008	8.84	1240.06337	228250912	2	10	36	0.9997389	Hex10dHex4H+2-T3	HexNAC2Hex3NeuAC2dHex5H+2-T1dHex-H2O
9.04	1259.02099	12636211	9.08	1269.08373	13407806	2	10	23	0.9992758	HexNAC3Hex3NeuAC1dHex6H+2-T3	HexNAC5Hex5NeuAC1dHex1H+2-T1Hex-H2O
7.68	930.90686	10514945	7.71	940.96995	10556602	2	10	22	0.9992514	HexNAC4Hex4dHex1H+2-T3	HexNAC4Hex3NeuAC1H+1NH4+1-T1Hex-H2O
8.82	1157.47813	23299374	8.85	1167.54104	25301754	2	10	31	0.9991861	HexNAC4Hex5NeuAC1dHex1H+2-T3	HexNAC7Hex1dHex3H+1K+1-T3dHex-H2O
7.90	1032.44689	8884180	7.94	1042.50966	8858641	2	10	22	0.9991705	HexNAC5Hex4dHex1H+2-T3	HexNAC3Hex3NeuAC1H+1NH4+1-T1Hex-H2O
9.78	1558.11511	22925302	9.81	1568.17854	24580040	2	10	30	0.9991136	HexNAC1Hex11NeuAC1dHex6H+2-T	HexNAC3Hex4NeuAC3dHex5H+2-T1NeuAC-H2O
8.80	1240.98430	10457655	8.84	1251.04883	11399541	2	10	17	0.9989141	HexNAC3Hex3NeuAC5H+2-T3	Hex10dHex4H+1Na+1-T3dHex-H2O
6.72	1470.63748	9545792	6.77	1490.76369	9189555	1	10	14	0.9987442	HexNAC2Hex5H+1-T3	HexNAC+HexNAC+Hex+Hex-4(H2O)
7.08	951.41852	2835175	7.12	961.48184	2805650.75	2	10	20	0.9986899	HexNAC5Hex3dHex1H+2-T3	HexNAC5Hex2NeuAC1H+1NH4+1-T1Hex-H2O
7.66	1860.80341	3366765	7.70	1880.92910	3340660.5	1	10	13	0.9986582	HexNAC4Hex4dHex1H+1-T3	HexNAC+HexNAC+Hex+Hex-4(H2O)
8.43	1084.45273	39776884	8.47	1094.51582	41829132	2	10	32	0.9985408	HexNAC2Hex3NeuAC1dHex5H+2-T3	HexNAC4Hex5NeuAC1H+2-T3dHex-H2O
7.79	1632.69132	6970936.5	7.84	1652.81755	6994956	1	10	13	0.9985286	HexNAC2Hex6H+1-T3	Hex+Hex+Hex+Hex-4(H2O)
7.46	1495.66646	168418.797	7.51	1515.79413	169358.516	1	5	0	0.9984599	HexNAC3Hex3dHex1H+1-T3	HexNAC+HexNAC+Hex+Hex-4(H2O)
7.89	982.91099	1423361.63	7.93	992.97446	1457163.38	2	10	26	0.9982689	HexNAC3Hex3NeuAC1H+2-T3	HexNAC6Hex1dHex2H+1K+1-T3Hex-H2O
9.79	1566.62682	14472047	9.82	1576.68956	15503985	2	10	34	0.9982379	HexNAC1Hex12dHex5H+2-T3	HexNAC3Hex5NeuAC2dHex6H+2-T1NeuAC-H2O
											dHex+Hex-2(H2O)

Figure 3.21 List Exported by 【Save Labelled Peak-Pairs】

Select Output → Save Peak with Glycone to open the file-save dialog and save the file in .xlsx format. The exported data consist of target peak information retained after 【MS/MS Glycone】 , including the retention times, m/z values, ion intensities of paired T-NCE10 precursor ions, and residue information matched to the mass pool. The results are shown in Figure 3.22.

RT (min)	m/z	Intensity	#Charge	#Score	#Glycone	dHex-H2O	Hex-H2O	dHex	Hex	HexNAC-H2O	NeuAC-H2O	dHex+dHex-2(H2O)
6.31	745.80501	282385.469	2	5	12	0	0	0	0	0	0	0
6.70	746.81290	2754944	2	10	12	0	0	0	0	0	0	0
7.76	827.83974	1768703.38	2	10	12	0	0	0	0	0	0	0
7.44	848.35301	386537.344	2	10	13	0	0	0	0	0	0	0
6.66	849.87813	2107797.25	2	10	22	0	3	1	3	1	0	0
7.10	857.87548	427515.375	2	10	16	0	2	0	0	1	0	0
6.65	860.86863	781349.875	2	10	12	0	0	0	0	0	0	0
7.10	868.86619	490158.531	2	10	12	0	0	0	0	0	0	0
6.51	878.38883	279681.594	2	10	13	0	1	0	0	0	0	0
7.17	901.88403	614266.625	2	10	20	0	1	0	0	0	1	0
8.71	906.38692	325134.531	2	0	19	0	5	0	0	1	0	0
7.18	912.87474	562838.125	2	10	14	0	0	0	0	0	2	0
8.43	918.38748	370731.563	2	10	14	0	3	0	0	0	0	0
7.66	930.71437	77389.0938	2	0	19	0	15	0	2	3	0	0
7.68	930.90686	10514945	2	10	22	0	22	0	2	5	0	0
8.11	938.90263	1802983.63	2	10	18	0	7	0	0	0	0	0
7.67	941.89494	2831779.5	2	10	12	0	0	0	0	0	0	0
7.72	949.88296	212594.813	2	10	12	0	0	0	0	0	0	0
8.13	949.89315	840292.5	2	10	12	0	0	0	0	0	0	0
7.08	951.41852	2835175	2	10	20	0	6	0	3	1	0	0
7.41	959.41554	562263.125	2	10	17	0	3	0	2	0	0	0

Figure 3.22 List Exported by 【Save Peak with Glycone】

3.10 Project Saving and Loading

Select File → Save Project to open the file-save dialog and save the project as a MATLAB workspace file (.mat). The project includes all processing parameters, intermediate results, and spectral data, facilitating later review or modification of the analysis workflow.

Select File → Load Project to open the file-selection dialog and import a project

in .mat format into the software.